

SOLIDWORKS API (.NET VBA C#) — Reference (327/337): View (part 7 of 10)

SOLIDWORKS API Documentation (converted from CHM)

IView (part 7 of 10) (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDimensionInfo3 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDimensionInfo3 Method (IView)

Obsolete. Superseded by [IView::GetDimensionInfo6](#) and [IView::IGetDimensionInfo6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDimensionInfo3() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As System.Double

value = instance.IGetDimensionInfo3()
```

C#

```
System.double IGetDimensionInfo3()
```

C++/CLI

```
System.double IGetDimensionInfo3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetDimensionInfo3](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDimensionInfo4 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDimensionInfo4 Method (IView)

Obsolete. Superseded by [IView::GetDimensionInfo6](#) and [IView::IGetDimensionInfo6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDimensionInfo4() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As System.Double

value = instance.IGetDimensionInfo4()
```

C#

```
System.double IGetDimensionInfo4()
```

C++/CLI

```
System.double IGetDimensionInfo4();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetDimensionInfo4](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDimensionInfo5 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDimensionInfo5 Method (IView)

Obsolete. Superseded by [IView::GetDimensionInfo6](#) and [IView::IGetDimensionInfo6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDimensionInfo5() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As System.Double

value = instance.IGetDimensionInfo5()
```

C#

```
System.double IGetDimensionInfo5()
```

C++/CLI

```
System.double IGetDimensionInfo5();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetDimensionInfo5](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDimensionInfo6 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDimensionInfo6 Method (IView)

Obsolete. Superseded by [IView::GetDimensionInfo7](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDimensionInfo6() As System.Double
```

Visual Basic (Usage)

```
Dim instance As IView
Dim value As System.Double

value = instance.IGetDimensionInfo6()
```

C#

```
System.double IGetDimensionInfo6()
```

C++/CLI

```
System.double IGetDimensionInfo6();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [View::IGetDimensionInfo6](#).

Remarks

The data returned is an array of doubles as follows:

```
[ dimensionCount, [ dimensionType, LineStyleIndex, LineWidth, Color, LayerID, LayerOverride, refPoint1a[3], refPoint1b[3],
refPoint2a[3], refPoint2b[3], refPoint3[3], textPoint[3], textDirection[3], dimNormal[3], angularDimInfo[9], precision, arrowLength,
arrowHeadWidth, arrowHeadStyle, witnessGap, witnessExt, dimValue, textHeight, arrowsOut, isDiameter,
RadialDimensionflags ] ]
```

where:

dimensionCount = number of dimensions found in the drawing view

For each dimension, you have a repeating set of the following data:

dimensionType = returns 0 = Linear, 1 = Angular, 2 = Radial, 3 = Ordinate

LineStyleIndex = line style; valid line styles as defined in [swLineStyles_e](#)

LineWidth = integer value defining the line width; valid width values as defined in [swLineWeights_e](#)

Color = COLORREF returned as an integer; return value could be 0 or -1 for default color

LayerID = integer value indicating which layer holds this entity; this integer value is the array index value into the layerList array; obtain the layerList array using [ILayerMgr::GetLayerList](#) or [ILayerMgr::IGetLayerList](#); a value of 1 indicates that this item is not on a layer

LayerOverride = integer with bit flags set to determine which properties, if any, have been overridden with respect to the [ILayer](#) default properties; if the bit value is set, then the specific property or properties have been overridden; the bit indicators are: color = 0x1, style = 0x2, and width = 0x4; therefore, if *LayerOverride* was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer

refPoint1a[3] = array of three doubles

refPoint1b[3] = array of three doubles

refPoint2a[3] = array of three doubles

refPoint2b[3] = array of three doubles

refPoint3[3] = array of three doubles

textPoint[3] = array of three doubles; this is the upper-left corner of the dimension bounding box

textDirection[3] = array of three doubles

dimNormal[3] = array of three doubles

angularDimInfo[9] This data is only returned for angular dimensions; the array of 9 values is as follows:

quadrant = as defined in [swQuadrant_e](#)

directionLine1[3] = array of three doubles

directionLine2[3] = array of three doubles

isInteriorAngle = BOOLEAN returned as a double; true if it is an interior angle

isFlipped = BOOLEAN returned as a double; true if it is flipped

precision = number of decimal places

arrowLength = arrow length

arrowHeadWidth = arrowhead width

arrowHeadHeight = arrowhead height

arrowHeadStyle = arrowhead style as defined in [swArrowStyle_e](#)

witnessGap = extension gap

witnessExt = extension's extension

dimValue = dimension value

textHeight = dimension text height

arrowsOut = BOOLEAN returned as a double and is true if the arrows are outside

isDiameter = BOOLEAN returned as a double and is used by radial dimensions; if this value is true, then the *dimValue* returned is the diameter value, if false, then the *dimValue* returned is the radial value

RadialDimensionflags = bit code defining properties for radial dimensions or 0 if the dimension is not radial. This element contains a bitwise OR of the following values:

RADIAL_INSIDE - Specifies that the arrows of the radial dimension are displayed inside the arc/circle.

RADIAL_2ND_ARROW - Specifies that there should be two outside arrows displayed with the radial dimension.

RADIAL_SOLID_LEADER - Specifies that the leader line should be drawn in solid.

RADIAL_CALLOUT - Specifies that there should be text displayed with the dimension.

RADIAL_IS_RADIAL - Specifies that the leader line should remain radial.

RADIAL_IS_BENT - Specifies that the leader line should be bent at the end.

RADIAL_IS_LINEAR - Specifies that the leader line should be straight at the end.

RADIAL_DOC_2ND_ARROW - Specifies that there should be two outside document arrows displayed with the radial dimension.

RADIAL_HAS_SNAPPED_DIR - Specifies that the arrow should snap to the grid.

The maximum number of doubles contained in the array will be (1 + *dimensionCount* * 51).

This method does not support [hole callouts](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetDimensionCount4 Method](#)
[IView::GetDimensionDisplayInfo5 Method](#)
[IView::GetDimensionDisplayInfoSize2 Method](#)
[IView::GetDimensionDisplayString4 Method](#)
[IView::GetDimensionIds4 Method](#)
[IView::GetDimensionInfo6 Method](#)
[IView::GetDimensionString4 Method](#)
[IView::GetFirstDisplayDimension5 Method](#)
[IView::IGetDimensionDisplayInfo5 Method](#)
[IView::IGetDimensionDisplayString4 Method](#)
[IView::IGetDimensionIds4 Method](#)
[IView::IGetDimensionString4 Method](#)
[IView::ProjectedDimensions Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDimensionString Method (IView)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IView Interface](#) : IGetDimensionString Method (IView)

Obsolete. Superseded by [IView::GetDimensionString4](#) and [IView::IGetDimensionString4](#).

.NET Syntax

Visual Basic (Declaration)

Function IGetDimensionString() As System.String

Visual Basic (Usage)

Dim instance As [IView](#)
Dim value As System.String



```
value = instance.IGetDimensionString()
```

C#

```
System.string IGetDimensionString()
```

C++/CLI

```
System.String^ IGetDimensionString();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetDimensionString](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDimensionString2 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDimensionString2 Method (IView)

Obsolete. Superseded by [IView::GetDimensionString4](#) and [IView::IGetDimensionString4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDimensionString2() As System.String
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.String
```

```
value = instance.IGetDimensionString2()
```

C#

```
System.string IGetDimensionString2()
```

C++/CLI

```
System.String^ IGetDimensionString2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetDimensionString2](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDimensionString3 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDimensionString3 Method (IView)

Obsolete. Superseded by [IView::GetDimensionString4](#) and [IView::IGetDimensionString4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDimensionString3() As System.String
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.String  
  
value = instance.IGetDimensionString3()
```

C#

```
System.string IGetDimensionString3()
```

C++/CLI

```
System.String^ IGetDimensionString3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetDimensionString3](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetDimensionString4 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDimensionString4 Method (IView)

Gets the strings containing the text associated with each dimension in the current drawing sheet or the current drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDimensionString4() As System.String
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.String  
  
value = instance.IGetDimensionString4()
```

C#

```
System.string IGetDimensionString4()
```

C++/CLI

```
System.String^ IGetDimensionString4();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of strings containing the text associated with each dimension

Visual Basic for Applications (VBA) Syntax

See [View::IGetDimensionString4](#).

Remarks

Use [IView::GetDimensionCount4](#) to determine the number of strings to be returned.

This method does not support [hole callouts](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetDimensionDisplayInfo5 Method](#)
[IView::GetDimensionDisplayInfoSize2 Method](#)
[IView::GetDimensionDisplayString4 Method](#)
[IView::GetDimensionIds4 Method](#)
[IView::GetDimensionInfo6 Method](#)
[IView::GetDimensionString4 Method](#)
[IView::GetFirstDisplayDimension5 Method](#)
[IView::IGetDimensionDisplayInfo5 Method](#)
[IView::IGetDimensionDisplayString4 Method](#)
[IView::IGetDimensionIds4 Method](#)
[IView::IGetDimensionInfo6 Method](#)
[IView::ProjectedDimensions Property](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDisplayData Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDisplayData Method (IView)

Obsolete. Superseded by [IView::GetDisplayData3](#) and [IView::IGetDisplayData3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDisplayData() As DisplayData
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As DisplayData

value = instance.IGetDisplayData()
```

C#

```
DisplayData IGetDisplayData()
```

C++/CLI

```
DisplayData^ IGetDisplayData();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetDisplayData](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDisplayData2 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDisplayData2 Method (IView)

Obsolete. Superseded by [IView::GetDisplayData3](#) and [IView::IGetDisplayData3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDisplayData2() As DisplayData
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As DisplayData  
  
value = instance.IGetDisplayData2()
```

C#

```
DisplayData IGetDisplayData2()
```

C++/CLI

```
DisplayData^ IGetDisplayData2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetDisplayData2](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDisplayData3 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDisplayData3 Method (IView)

Gets the the [DisplayData](#) object for this drawing view containing only those model items that are visible in the view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDisplayData3() As DisplayData
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As DisplayData  
  
value = instance.IGetDisplayData3()
```

C#

```
DisplayData IGetDisplayData3()
```

C++/CLI

```
DisplayData^ IGetDisplayData3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IDisplayData](#) object

Visual Basic for Applications (VBA) Syntax

See [View::IGetDisplayData3](#).

Remarks

The data kept with the [IDisplayData](#) object is strictly for display purposes and allows you to recreate unintelligent geometry. It does not provide associations or detailed information about the geometry.

This method supersedes the now obsolete [View::IGetDisplayData2](#) and should be used in its place. The previous version of this method returned model items that were not imported to the drawing view and, therefore, not visible in the drawing view.

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetDisplayData3 Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDisplayDimensions Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDisplayDimensions Method (IView)

Gets all of the display dimensions on this drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDisplayDimensions( _  
    ByVal NumDisplayDim As System.Integer _  
) As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim NumDisplayDim As System.Integer  
Dim value As DisplayDimension  
  
value = instance.IGetDisplayDimensions(NumDisplayDim)
```

C#

```
DisplayDimension IGetDisplayDimensions(  
    System.int NumDisplayDim  
)
```

C++/CLI

```
DisplayDimension^ IGetDisplayDimensions(  
& System.int NumDisplayDim  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumDisplayDim
Total number of display dimensions on this drawing view

Return Value

- in-process, unmanaged C++: Pointer to an array of [display dimensions](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Use this method to obtain the array of display dimensions all at once instead of calling [IView::GetFirstDisplayDimension5](#) and then repeatedly calling [IDisplayDimension::GetNext5](#) to obtain the remaining display dimensions on this drawing view.

Before calling this method, call [IView::GetDisplayDimensionCount](#) to get the value for numDisplayDim.

NOTE: Display dimensions are not the same as [actual model dimensions](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetDisplayDimensions Method](#)

Availability

SOLIDWORKS 2009 SP1, Revision Number 17.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDowelSymbols Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetDowelSymbols Method (IView)

Gets all of the dowel symbols on this drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetDowelSymbols( _  
    ByVal NumDowelSymbol As System.Integer _  
) As DowelSymbol
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim NumDowelSymbol As System.Integer  
Dim value As DowelSymbol  
  
value = instance.IGetDowelSymbols(NumDowelSymbol)
```

C#

```
DowelSymbol IGetDowelSymbols(  
    System.int NumDowelSymbol  
)
```

C++/CLI

```
DowelSymbol^ IGetDowelSymbols(  
& System.int NumDowelSymbol  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumDowelSymbol

Total number of dowel symbols on this drawing view

Return Value

- in-process, unmanaged C++: Pointer to an array of [dowel symbols](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Use this method to obtain the array of dowel symbols all at once instead of calling [IView::GetFirstDowelSymbol](#) and then repeatedly calling [IDowelSymbol::GetNext](#) to obtain the remaining dowel symbols on this drawing view.

Before calling this method, call [IView::GetDowelSymbolCount](#) to get the value for numDowelSymbol.

 See Also

[IView Interface](#)
[IView Members](#)
[IView::GetDowelSymbols Method](#)

 Availability

SOLIDWORKS 2009 SP1, Revision Number 17.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEllipses Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetEllipses Method (IView)

Obsolete. Superseded by [IView::GetEllipses5](#) and [IView::IGetEllipses5](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetEllipses() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As System.Double

value = instance.IGetEllipses()
```

C#

```
System.double IGetEllipses()
```

C++/CLI

```
System.double IGetEllipses();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [VView::IGetEllipses](#).

 See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEllipses2 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetEllipses2 Method (IView)

Obsolete. Superseded by [IView::GetEllipses5](#) and [IView::IGetEllipses5](#).

 .NET Syntax

Visual Basic (Declaration)

Function IGetEllipses2() As System.Double

Visual Basic (Usage)



```
Dim instance As IView
Dim value As System.Double

value = instance.IGetEllipses2()
```

C#

```
System.double IGetEllipses2()
```

C++/CLI

```
System.double IGetEllipses2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetEllipses2](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetEllipses3 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetEllipses3 Method (IView)

Obsolete. Superseded by [IView::GetEllipses5](#) and [IView::IGetEllipses5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEllipses3() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As System.Double

value = instance.IGetEllipses3()
```

C#

```
System.double IGetEllipses3()
```

C++/CLI

```
System.double IGetEllipses3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetEllipses3](#).

See Also

[IView Interface](#)
[IView Members](#)

IGetEllipses4 Method (IView)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetEllipses4 Method (IView)

Obsolete. Superseded by [IView::GetEllipses5](#) and [IView::IGetEllipses5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEllipses4() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double  
  
value = instance.IGetEllipses4()
```

C#

```
System.double IGetEllipses4()
```

C++/CLI

```
System.double IGetEllipses4();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetEllipses4](#).

See Also

[IView Interface](#)
[IView Members](#)

IGetEllipses5 Method (IView)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetEllipses5 Method (IView)

Gets all of the ellipses added by a user in this drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEllipses5() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double  
  
value = instance.IGetEllipses5()
```

C#

```
System.double IGetEllipses5()
```

C++/CLI

```
System.double IGetEllipses5();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (**Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

This method only returns ellipses that were sketched by a user in this drawing view. Use [IView::GetPolylines6](#) or [IView::GetPolylines6](#) to access the solid model's projected display data in the drawing view.

All part and assembly geometry shown in a drawing view is represented by polylines. To access the polylines, use [View::GetPolylines3](#). The return values are in an array of doubles:

```
[ Color, LineType, LineStyleIndex, LineWidth, LayerID, LayerOverride, StartPt[3], EndPt[3], CenterPt[3], MajorPt[3], MinorPt[3], Direction .. ]
```

where:

Color = [COLORREF](#) returned as an integer. Return value can be 0 or -1 for default color.

LineType = line type. Valid returns as defined in [swLineTypes e](#). A *lineType* is a combination of a *lineStyle* and *lineWeight*.

LineStyleIndex = line style. Valid line styles as defined in [swLineStyles e](#).

LineWidth = integer value defining the line width. Valid width values as defined in [swLineWeights e](#).

LayerID = integer value indicating that layer holds this entity. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::!GetLayerById](#).

LayerOverride = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if *LayerOverride* is returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

StartPt[3] = array of 3 doubles (X,Y,Z) describing the ellipse start point.

EndPt[3] = array of 3 doubles (X,Y,Z) describing the ellipse end point. If the ellipse is closed, then this will be the same point as the *StartPt*.

CenterPt[3] = array of 3 doubles (X,Y,Z) describing the ellipse center point.

MajorPt[3] = array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the major axis.

MinorPt[3] = array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the minor axis.

Direction = -1 for clockwise, +1 for counterclockwise.

This set of data repeats for each ellipse in the view. The size of the array is (NumEllipses * 22). To determine the number of ellipses, use [IView::GetEllipseCount](#).

The data returned from this method is in terms of view space. If you want the data in terms of sheet space (for example, the 0,0 origin being the lower-left corner of the sheet), then combine this data with the three return values from [IView::GetXForm](#) or [IView::!GetXForm](#).

The sheet must be visible. See [ISheet::SheetFormatVisible](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetEllipses5 Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFaceHatches Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFaceHatches Method (IView)

Gets the face hatches in the view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFaceHatches( _  
    ByVal NumFacesHatches As System.Integer _  
) As FaceHatch
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim NumFacesHatches As System.Integer  
Dim value As FaceHatch  
  
value = instance.IGetFaceHatches(NumFacesHatches)
```

C#

```
FaceHatch IGetFaceHatches(  
    System.int NumFacesHatches  
)
```

C++/CLI

```
FaceHatch^ IGetFaceHatches(  
&    System.int NumFacesHatches  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumFacesHatches

Number of face hatches in the view

Return Value

- in-process, unmanaged C++: Pointer to an array of [face hatches](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Call [IView::GetFaceHatchCount](#) before calling this method to get the value for NumFacesHatches.

To get the number of solid-fill hatches in a detail, broken, or crop view, use [IView::GetSolidHatchCount](#). To get solid hatches in detail, broken, or crop views, use [IView::GetSolidHatchInfo](#) or [IView::IGetSolidHatchInfo](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetFaceHatches Method](#)
[IView::ScaleHatchPattern Property](#)
[IModelDoc2::InsertHatchedFace Method](#)
[IView::ScaleHatchPattern Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFirstAnnotation Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstAnnotation Method (IView)

Obsolete. Superseded by [IView::GetFirstAnnotation3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstAnnotation() As Annotation
```

Visual Basic (Usage)

```
Dim instance As IView  
Dim value As Annotation  
  
value = instance.IGetFirstAnnotation()
```

C#

```
Annotation IGetFirstAnnotation()
```

C++/CLI

```
Annotation^ IGetFirstAnnotation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstAnnotation](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstAnnotation2 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstAnnotation2 Method (IView)

Obsolete. Superseded by [IView::GetFirstAnnotation3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstAnnotation2() As Annotation
```

Visual Basic (Usage)

```
Dim instance As IView  
Dim value As Annotation  
  
value = instance.IGetFirstAnnotation2()
```

C#

```
Annotation IGetFirstAnnotation2()
```

C++/CLI

```
Annotation^ IGetFirstAnnotation2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstAnnotation2](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstCenterOfMass Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstCenterOfMass Method (IView)

Gets the first center of mass in this view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstCenterOfMass() As CenterOfMass
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As CenterOfMass  
  
value = instance.IGetFirstCenterOfMass()
```

C#

```
CenterOfMass IGetFirstCenterOfMass()
```

C++/CLI

```
CenterOfMass^ IGetFirstCenterOfMass();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to [ICenterOfMass](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetFirstCenterOfMass Method](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFirstCThread Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstCThread Method (IView)

Gets the first cosmetic thread in the view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstCThread() As CThread
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As CThread  
  
value = instance.IGetFirstCThread()
```

C#

```
CThread IGetFirstCThread()
```

C++/CLI

```
CThread^ IGetFirstCThread();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [cosmetic thread](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstCThread](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetFirstCThread Method](#)
[CThread::GetNext Method](#)
[CThread::IGetNext Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFirstCustomSymbol Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstCustomSymbol Method (IView)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstCustomSymbol() As CustomSymbol
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As CustomSymbol  
  
value = instance.IGetFirstCustomSymbol()
```

C#

```
CustomSymbol IGetFirstCustomSymbol()
```

C++/CLI

```
CustomSymbol^ IGetFirstCustomSymbol();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstCustomSymbol](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFirstDatumTag Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstDatumTag Method (IView)

Gets the first datum tag in the view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstDatumTag() As DatumTag
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As DatumTag  
  
value = instance.IGetFirstDatumTag()
```

C#

```
DatumTag IGetFirstDatumTag()
```

C++/CLI

```
DatumTag^ IGetFirstDatumTag();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [datum tag](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstDatumTag](#).

See Also

[IView Interface](#)
[IView Members](#)
[IDatumTag::IGetNext Method](#)
[IDatumTag::GetNext Method](#)
[IView::GetFirstDatumTag Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFirstDatumTargetSym Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstDatumTargetSym Method (IView)

Gets the first datum target symbol in the view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstDatumTargetSym() As DatumTargetSym
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As DatumTargetSym  
  
value = instance.IGetFirstDatumTargetSym()
```

C#

```
DatumTargetSym IGetFirstDatumTargetSym()
```

C++/CLI

```
DatumTargetSym^ IGetFirstDatumTargetSym();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [datum target symbol](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstDatumTargetSym](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFirstDisplayDimension Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstDisplayDimension Method (IView)

Obsolete. Superseded by [IView::GetFirstDisplayDimension5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstDisplayDimension() As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As DisplayDimension  
  
value = instance.IGetFirstDisplayDimension()
```

C#

```
DisplayDimension IGetFirstDisplayDimension()
```

C++/CLI

```
DisplayDimension^ IGetFirstDisplayDimension();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstDisplayDimension](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFirstDisplayDimension2 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstDisplayDimension2 Method (IView)

Obsolete. Superseded by [IView::GetFirstDisplayDimension5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstDisplayDimension2() As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As DisplayDimension
```

```
value = instance.IGetFirstDisplayDimension2()
```

C#

```
DisplayDimension IGetFirstDisplayDimension2()
```

C++/CLI

```
DisplayDimension^ IGetFirstDisplayDimension2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstDisplayDimension2](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstDisplayDimension3 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstDisplayDimension3 Method (IView)

Obsolete. Superseded by [IView::GetFirstDisplayDimension5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstDisplayDimension3() As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As DisplayDimension  
  
value = instance.IGetFirstDisplayDimension3()
```

C#

```
DisplayDimension IGetFirstDisplayDimension3()
```

C++/CLI

```
DisplayDimension^ IGetFirstDisplayDimension3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstDisplayDimension3](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstDowelSymbol Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstDowelSymbol Method (IView)

Gets the first dowel symbol in the view.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetFirstDowelSymbol() As DowelSymbol
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As DowelSymbol  
  
value = instance.IGetFirstDowelSymbol()
```

C#

```
DowelSymbol IGetFirstDowelSymbol()
```

C++/CLI

```
DowelSymbol^ IGetFirstDowelSymbol();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [dowel symbol](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstDowelSymbol](#).

See Also

[IView Interface](#)
[IView Members](#)
[IDowelSymbol::GetNext Method](#)
[IDowelSymbol::IGetNext Method](#)
[IView::GetFirstDowelSymbol Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFirstGTOL Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstGTOL Method (IView)

Gets the first geometric tolerance in this view.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetFirstGTOL() As Gtol
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As Gtol  
  
value = instance.IGetFirstGTOL()
```

C#

```
Gtol IGetFirstGTOL()
```

C++/CLI

```
Gtol^ IGetFirstGTOL();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [GToI](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstGTOL](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetFirstGTOL Method](#)
[IGtol::GetNextGTOL Method](#)
[IGtol::IGetNextGTOL Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetFirstMultiJogLeader Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstMultiJogLeader Method (IView)

Gets the first multi-jog leader in the view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstMultiJogLeader() As MultiJogLeader
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As MultiJogLeader  
  
value = instance.IGetFirstMultiJogLeader()
```

C#

```
MultiJogLeader IGetFirstMultiJogLeader()
```

C++/CLI

```
MultiJogLeader^ IGetFirstMultiJogLeader();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [multi-jog leader](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstMultiJogLeader](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetFirstMultiJogLeader Method](#)
[IMultiJogLeader::GetNext Method](#)
[IMultiJogLeader::IGetNext Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

IGetFirstNote Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstNote Method (IView)

Gets the first note in the view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstNote() As Note
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As Note

value = instance.IGetFirstNote()
```

C#

```
Note IGetFirstNote()
```

C++/CLI

```
Note^ IGetFirstNote();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [note](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstNote](#).

Remarks

The sheet must be visible. See [ISheet::SheetFormatVisible](#).

See Also

[IView Interface](#)
[IView Members](#)
[INote::GetNext Method](#)
[INote::IGetNext Method](#)
[IView::GetFirstNote Method](#)

IGetFirstRevisionCloud Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstRevisionCloud Method (IView)

Gets the first revision cloud annotation in this view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstRevisionCloud() As RevisionCloud
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As RevisionCloud
```

```
value = instance.IGetFirstRevisionCloud()
```

C#

```
RevisionCloud IGetFirstRevisionCloud()
```

C++/CLI

```
RevisionCloud^ IGetFirstRevisionCloud();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an [IRevisionCloud](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetFirstRevisionCloud Method \(\)](#)
[IView::GetRevisionCloudCount Method \(\)](#)
[IView::GetRevisionClouds Method \(\)](#)
[IView::IGetRevisionClouds Method \(\)](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstSFSymbol Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstSFSymbol Method (IView)

Gets the first surface-finish symbols in the view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstSFSymbol() As SFSymbol
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As SFSymbol  
  
value = instance.IGetFirstSFSymbol()
```

C#

```
SFSymbol IGetFirstSFSymbol()
```

C++/CLI

```
SFSymbol^ IGetFirstSFSymbol();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [surface-finish symbol](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstSFSymbol](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetFirstSFSymbol Method](#)
[ISFSymbol::GetNext Method](#)
[ISFSymbol::IGetNext Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFirstWeldSymbol Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetFirstWeldSymbol Method (IView)

Gets the first weld symbol in the view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFirstWeldSymbol() As WeldSymbol
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As WeldSymbol  
  
value = instance.IGetFirstWeldSymbol()
```

C#

```
WeldSymbol IGetFirstWeldSymbol()
```

C++/CLI

```
WeldSymbol^ IGetFirstWeldSymbol();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [weld symbol](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetFirstWeldSymbol](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetFirstWeldSymbol Method](#)
[IWeldSymbol::GetNext Method](#)
[IWeldSymbol::IGetNext Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetGTols Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetGTols Method (IView)

Gets all of the geometric tolerances on this drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetGTols(  
    ByVal NumGTol As System.Integer  
) As Gtol
```

Visual Basic (Usage)



```
Dim instance As IView
Dim NumGTol As System.Integer
Dim value As Gtol

value = instance.IGetGTols(NumGTol)
```

C#

```
Gtol IGetGTols(
    System.int NumGTol
)
```

C++/CLI

```
Gtol^ IGetGTols(
    & System.int NumGTol
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumGTol

Total number of geometric tolerances on this drawing view.

Return Value

- in-process, unmanaged C++: Pointer to an array of [geometric tolerances](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Use this method to obtain the array of geometric tolerances all at once instead of calling [IView::GetFirstGTOL](#) and then repeatedly calling [IGtol::GetNextGTOL](#) to obtain the remaining geometric tolerances in the drawing view.

Before calling this method, call [IView::GetGTolCount](#) to get the value for numGTol.

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetGTols Method](#)

Availability

SOLIDWORKS 2009 SP1, Revision Number 17.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetLines Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetLines Method (IView)

Obsolete. Superseded by [IView::GetLines4](#) and [IView::IGetLines4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLines() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As System.Double

value = instance.IGetLines()
```

C#

```
System.double IGetLines()
```

C++/CLI

```
System.double IGetLines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetLines](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetLines2 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetLines2 Method (IView)

Obsolete. Superseded by [IView::GetLines4](#) and [IView::IGetLines4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLines2() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double  
  
value = instance.IGetLines2()
```

C#

```
System.double IGetLines2()
```

C++/CLI

```
System.double IGetLines2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetLines2](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetLines3 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetLines3 Method (IView)

Obsolete. Superseded by [IView::GetLines4](#) and [IView::IGetLines4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLines3() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double  
  
value = instance.IGetLines3()
```

C#

```
System.double IGetLines3()
```

C++/CLI

```
System.double IGetLines3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetLines3](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetLines4 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetLines4 Method (IView)

Gets all of the lines in the view with an option to include or exclude crosshatch lines.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetLines4( _  
    ByVal CrossHatchOption As System.Short, _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim CrossHatchOption As System.Short  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetLines4(CrossHatchOption, ArraySize)
```

C#

```
System.double IGetLines4(  
    System.short CrossHatchOption,  
    System.int ArraySize  
)
```

C++/CLI

```
System.double IGetLines4(  
& System.short CrossHatchOption,  
& System.int ArraySize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CrossHatchOption

Crosshatch option as defined in [swCrossHatchFilter_e](#)

ArraySize

Number of lines in the view

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This method only returns lines that were sketched in this drawing view. Use [IView::GetPolylines6](#) or [IView::IGetPolylines6](#) to access the solid model's projected display data in this view.

Call [IView::GetLineCount2](#) before calling this method to get the value for *ArraySize*.

The return value is the following array of doubles:

[*Color*, *LineType*, *LineStyleIndex*, *LineWidth*, *LayerID*, *LayerOverride*, *StartPt*[3], *EndPt*[3], ...]

where all data values are returned as doubles:

- *Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- *LineType* = line type. Valid returns as defined in [swLineTypes_e](#). A *lineType* is a combination of a *lineStyle* and *lineWeight*.
- *LineStyleIndex* = line style. Valid line styles as defined in [swLineStyles_e](#).
- *LineWidth* = integer value defining the line width. Valid width values as defined in [swLineWeights_e](#).
- *LayerID* = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).
- *LayerOverride* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style : 0x2, and width = 0x4. Therefore, if *LayerOverride* is returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
- *StartPt*[3] = array of 3 doubles (X,Y,Z) describing the line start point.
- *EndPt*[3] = array of 3 doubles (X,Y,Z) describing the line end point.

This set of data repeats for each line in the view. The number of doubles returned is (lineCount * 12).

The data returned from this method is in terms of view space. If you want the data in terms of sheet space (that is, the 0,0 origin being the lower-left corner of the sheet), then combine this data with the three return values from [IView::GetXForm](#) or [IView::IGetXForm](#).

The sheet must be visible. See [ISheet::SheetFormatVisible](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetLines4 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMultiJogLeaders Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetMultiJogLeaders Method (IView)

Gets all of the multi-jog leaders in this drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMultiJogLeaders(  
    ByVal NumMultiJogLeader As System.Integer _  
) As MultiJogLeader
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim NumMultiJogLeader As System.Integer  
Dim value As MultiJogLeader  
  
value = instance.IGetMultiJogLeaders(NumMultiJogLeader)
```

C#

```
MultiJogLeader IGetMultiJogLeaders(  
    System.int NumMultiJogLeader  
)
```

C++/CLI

```
MultiJogLeader^ IGetMultiJogLeaders(  
&    System.int NumMultiJogLeader  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumMultiJogLeader

Number of multi-jog leaders in this drawing view.

Return Value

- in-process, unmanaged C++: Pointer to an array of [multi-jog leaders](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

Use this method to obtain the array of multi-jog leaders all at once instead of calling [IView::GetFirstMultiJogLeader](#) and then repeatedly calling [IMultiJogLeader::GetNext](#) to obtain the remaining multi-jog leaders in the drawing view.

Before calling this method, call [IView::GetMultiJogLeaderCount](#) to get the value for numMultiJogLeader.

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetMultiJogLeaders Method](#)

Availability

SOLIDWORKS 2009 SP1, Revision Number 17.1

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetNextView Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetNextView Method (IView)

Gets the next drawing view in the drawing.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNextView() As View
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As View

value = instance.IGetNextView()
```

C#

```
View IGetNextView()
```

C++/CLI

```
View^ IGetNextView();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Next [view](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetNextView](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetBaseView Method](#)
[IView::GetNextView Method](#)
[IView::IGetBaseView Method](#)
[IDrawingDoc::GetFirstView Method](#)
[IDrawingDoc::IGetFirstView Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetNotes Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetNotes Method (IView)

Gets the notes in this drawing view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNotes( _
    ByVal NumberOfNotes As System.Integer _
) As Note
```

Visual Basic (Usage)



```
Dim instance As IView
Dim NumberOfNotes As System.Integer
Dim value As Note

value = instance.IGetNotes(NumberOfNotes)
```

C#

```
Note IGetNotes(
    System.int NumberOfNotes
)
```

C++/CLI

```
Note^ IGetNotes(
    & System.int NumberOfNotes
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumberOfNotes
Number of notes in this drawing view

Return Value

- in-process, unmanaged C++: Pointer to an array of [notes](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Use this method to obtain the array of notes all at once instead of calling [IView::GetFirstNote](#) and then repeatedly calling [INote::GetNext](#) to obtain the remaining notes on this drawing view.

Call [IView::GetNoteCount](#) before calling this method to get NumberOfNotes.

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetNotes Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetOutline Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetOutline Method (IView)

Gets the bounding box for a view (sheet or drawing) in meters on the drawing page.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetOutline() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView
Dim value As System.Double

value = instance.IGetOutline()
```

C#

```
System.double IGetOutline()
```

C++/CLI

```
System.double IGetOutline();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [View::IGetOutline](#).

Remarks

The return value is an array of 4 doubles with the following format:

- 0, X min
- 1, Y min

- 2, X max
- 3, Y max

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetOutline Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetParabolas Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetParabolas Method (IView)

Obsolete. Superseded by [IView::GetParabolas2](#) and [IView::IGetParabolas2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetParabolas() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double  
  
value = instance.IGetParabolas()
```

C#

```
System.double IGetParabolas()
```

C++/CLI

```
System.double IGetParabolas();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetParabolas](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetParabolas2 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetParabolas2 Method (IView)

Gets all of the information about all of the parabolas in this view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetParabolas2() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double
```

```
value = instance.IGetParabolas2()
```

C#

```
System.double IGetParabolas2()
```

C++/CLI

```
System.double IGetParabolas2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return values are in an array of doubles:

[*Color*, *LineType*, *LineFont*, *LineWidth*, *LayerID*, *LayerOverride*, *StartPt*[3], *EndPt*[3], *FocusPt*[3], *ApexPt*[3]]

where:

Color = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

LineType = line type. Valid returns as defined in [swLineTypes.e](#). A *LineType* is a combination of a *LineStyle* and *LineWeight*.

LineFont = Value is used for polyline font information.

LineWidth = integer value defining the line width. Valid width values as defined in [swLineWeights.e](#).

LayerID = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).

LayerOverride = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if *LayerOverride* was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

StartPt[3] = array of 3 doubles (X,Y,Z) describing the parabola start point.

EndPt[3] = array of 3 doubles (X,Y,Z) describing the parabola end point.

FocusPt[3] = array of 3 doubles (X,Y,Z) describing the parabola focus point.

ApexPt[3] = array of 3 doubles (X,Y,Z) describing the parabola apex point.

This set of data repeats for each parabola in the view. The size of the array is (NumParabolas * 18). To determine the number of parabolas, use [IView::GetParabolaCount](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetParabolas2 Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPolylines Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetPolylines Method (IView)

Obsolete. Superseded by [IView::GetPolyLines6](#) and [IView::IGetPolyLines6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolylines() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double  
  
value = instance.IGetPolylines()
```

C#

```
System.double IGetPolylines()
```

C++/CLI

```
System.double IGetPolylines();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetPolylines](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetPolylines2 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetPolylines2 Method (IView)

Obsolete. Superseded by [IView::GetPolyLines6](#) and [IView::IGetPolyLines6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolylines2() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double  
  
value = instance.IGetPolylines2()
```

C#

```
System.double IGetPolylines2()
```

C++/CLI

```
System.double IGetPolylines2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetPolylines2](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetPolylines3 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetPolylines3 Method (IView)

Obsolete. Superseded by [IView::GetPolyLines6](#) and [IView::IGetPolyLines6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolylines3() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double  
  
value = instance.IGetPolylines3()
```

C#

```
System.double IGetPolylines3()
```

C++/CLI

```
System.double IGetPolylines3();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetPolylines3](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPolylines4 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetPolylines4 Method (IView)

Obsolete. Superseded by [IView::GetPolyLines6](#) and [IView::IGetPolyLines6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolylines4() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim value As System.Double  
  
value = instance.IGetPolylines4()
```

C#

```
System.double IGetPolylines4()
```

C++/CLI

```
System.double IGetPolylines4();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [View::IGetPolylines4](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetPolylines5 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetPolylines5 Method (IView)

Obsolete. Superseded by [IView::GetPolyLines6](#) and [IView::IGetPolyLines6](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolylines5( _  
    ByVal CrossHatchOption As System.Short, _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim CrossHatchOption As System.Short  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetPolylines5(CrossHatchOption, ArraySize)
```

C#

```
System.double IGetPolylines5(  
    System.short CrossHatchOption,  
    System.int ArraySize  
)
```

C++/CLI

```
System.double IGetPolylines5(  
& System.short CrossHatchOption,  
& System.int ArraySize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CrossHatchOption

ArraySize

Visual Basic for Applications (VBA) Syntax

See [View::IGetPolylines5](#).

See Also

[IView Interface](#)
[IView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetPolylines6 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetPolylines6 Method (IView)

Obsolete. Superseded by [IView::IGetPolylines7](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolylines6( _  
    ByVal CrossHatchOption As System.Short, _  
    ByVal ArraySize As System.Integer, _  
    ByRef Polylines As System.Double, _  
    ByVal EdgeArraySize As System.Integer _  
) As Edge
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim CrossHatchOption As System.Short  
Dim ArraySize As System.Integer  
Dim Polylines As System.Double  
Dim EdgeArraySize As System.Integer  
Dim value As Edge  
  
value = instance.IGetPolylines6(CrossHatchOption, ArraySize, Polylines, EdgeArraySize)
```

C#

```
Edge IGetPolylines6(  
    System.short CrossHatchOption,  
    System.int ArraySize,  
    out System.double Polylines,  
    System.int EdgeArraySize  
)
```

C++/CLI

```
Edge^ IGetPolylines6(  
& System.short CrossHatchOption,  
& System.int ArraySize,  
& [Out] System.double Polylines,  
& System.int EdgeArraySize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

- CrossHatchOption*
 - 0 = include crosshatch lines
 - 1 = exclude crosshatch lines
 - 2 = include only crosshatch lines
- ArraySize*
Size of array of lines
- Polylines*
Array of doubles representing the lines in the view (see **Remarks**)
- EdgeArraySize*
Size of array of modeling edges

Return Value

Array of modeling [edges](#) corresponding to the polylines in the view

Visual Basic for Applications (VBA) Syntax

See [View::IGetPolylines6](#).

Remarks

The difference between this method and [IView::GetPolyLinesAndCurves](#) and [IView::IGetPolyLinesAndCurves](#) is that splines and ellipses returned by these methods are returned as equation parameters, which are more precise definitions.

This method returns data for all visible solid model edges in the drawing view. This includes these types of edges:

- arcs: Returned using a center point, radius, and start and stop location.
- circles: Returned using a center point, radius, and start and stop location.
- ellipses: Returned as tessellated polylines.
- splines: Returned as tessellated polylines.
- straight lines: Returned as tessellated polylines, but because the edges are straight, there is no loss of data in the approximation.

NOTE: If the view contains silhouette edges, then the polylines that render them cannot correspond to an edge because an edge does not actually exist. For example, if the third and fourth polyline in the set of polylines returned describe a silhouette edge, then array positions 4 and 5 in the edge array will be null.

This method also lets you include or exclude crosshatch lines. This method does not return data for sketch segments added by a user in the drawing view.

To get...	Call...
Size of array needed to hold this data	IView::GetPolylineCount5
Items that were sketched in this drawing view	IView::GetArcs4 and IView::IGetArcs4 IView::GetLines4 and IView::IGetLines4 IView::GetSplines3 and IView::IGetSplines3 IView::GetEllipses5 and IView::IGetEllipses5

Format of the return value is the following array of doubles:

[*Type*, *GeomDataSize*, *GeomData[]*, *LineColor*, *LineStyle*, *LineFont*, *LineWeight*, *LayerID*, *LayerOverride*, *NumPolyPoints*, [x,y,z]]

where:

Type = underlying geometry type, possible values are:

- 0 Polyline type
- 1 Arc or Circle type

GeomDataSize = number of elements in the *GeomData* array, for *Type* = 0 this will be 0.

GeomData[] = geometric data that can be used to represent the underlying geometry type. The data returned in this array is based on the underlying geometry type:

- *Type* = 0. This array is empty
- *Type* = 1. [*centerPtX*, *centerPtY*, *centerPtZ*, *startPtX*, *startPtY*, *startPtZ*, *endPtX*, *endPtY*, *endPtZ*, *normalX*, *normalY*, *normalZ*]

LineColor = polyline color. This value is determined either by the explicitly set value or by the layer that the polyline is on.

LineStyle = value combines polyline font and weight information. This value can be used as an input to *GetLineFontInfo* and *GetLineFontName*. If this value is -1, then the user has probably modified the line display manually in the drawing and you should use the *LineFont* and *LineWeight* return values to recreate the exact polyline style.

LineFont = value is used for polyline font information. This value can be used as an input to the *GetLineFontInfo2* and *GetLineFontName2* functions. This value will only be valid if *LineStyle* is -1.

LineWeight = polyline weight where Thin = 0.0, Normal = 1.0, Thick = 2.0. This value will only be valid if *LineStyle* is -1.

LayerID = integer value indicating which layer holds this polyline. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) and [ILayerMgr::IGetLayerById](#).

LayerOverride = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if *LayerOverride* was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

NumPolyPoints = number of XYZ points found in the [x,y,z] return value

[x,y,z] = array of points used to describe the polyline. This array has *NumPolyPoints* points. This data will be returned for every polyline regardless of *Type*.

If...	Then...
Display mode of the view is: • Shaded • Shaded with Edges • Draft Quality • Fast HLR/HLV	A value is not returned. Use IView::SetDisplayMode3 to change Shaded or Shaded with Edges mode to Wireframe, Hidden Lines Removed (HLR), or Hidden Lines Visible (HLV), and then get the polylines.
Changes are made to the parts or assemblies shown in this drawing	Polylines are only generated that are in the visible viewing bounds when the drawing is opened.

Drawing is already open

All polylines in the drawing are generated. If you open a drawing that is zoomed in to a particular region, then the polylines that are outside the zoomed region do not exist if the parts or assemblies shown in this drawing have been changed. To force the generation of all possible polyline data, call [IModelDoc2::ViewZoomtofit2](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetDisplayMode2 Method](#)
[IView::GetPolylines6 Method](#)
[IView::GetPolyLinesAndCurves Method](#)
[IView::IGetPolyLinesAndCurves Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPolylines7 Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetPolylines7 Method (IView)

Gets all of the polylines in the view with an option to include or exclude crosshatch lines

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolylines7( _  
    ByVal CrossHatchOption As System.Short, _  
    ByVal ArraySize As System.Integer, _  
    ByRef Polylines As System.Double, _  
    ByVal EdgeArraySize As System.Integer _  
    ) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim CrossHatchOption As System.Short  
Dim ArraySize As System.Integer  
Dim Polylines As System.Double  
Dim EdgeArraySize As System.Integer  
Dim value As System.Object  
  
value = instance.IGetPolylines7(CrossHatchOption, ArraySize, Polylines, EdgeArraySize)
```

C#

```
System.object IGetPolylines7(  
    System.short CrossHatchOption,  
    System.int ArraySize,  
    out System.double Polylines,  
    System.int EdgeArraySize  
)
```

C++/CLI

```
System.Object^ IGetPolylines7(  
& System.short CrossHatchOption,  
& System.int ArraySize,  
& [Out] System.double Polylines,  
& System.int EdgeArraySize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CrossHatchOption

- 0 = include crosshatch lines
- 1 = exclude crosshatch lines
- 2 = include only crosshatch lines

ArraySize

Number of lines in the view

Polylines

Array of doubles of the lines in the view of size *ArraySize*

EdgeArraySize

Size of array of modeling edges

Return Value

- in-process, unmanaged C++: Pointer to an array of modeling [edges](#) and [silhouette edges](#) corresponding to polylines in the view
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

The difference between this method and [IView::GetPolyLinesAndCurves](#) and [IView::!GetPolyLinesAndCurves](#) is that splines and ellipses returned by these methods are returned as equation parameters, which are more precise definitions.

This method returns data for all visible solid model edges in the drawing view. This includes these types of edges:

- 1 = arcs: Returned using a center point, radius, and start and stop location.
- 1 = circles: Returned using a center point, radius, and start and stop location.
- 2 = ellipses: Returned as tessellated polylines.
- 3 = splines: Returned as tessellated polylines.
- 0 = straight lines: Returned as tessellated polylines, but because the edges are straight, there is no loss of data in the approximation.

For each arc:

- first double gives the type
- next double gives the number of doubles required to hold arc geometry (center, startPt, endPt, normal) == 12
- next three doubles give the coordinates of center
- next three doubles give the coordinates of startPt
- next three doubles give the coordinates of endPt
- next three doubles give the coordinates of the normal
- next six doubles are line attributes:
 - *LineColor* = polyline color. This value is determined either by the explicitly set value or by the layer that the polyline is on.
 - *LineStyle* = value combines polyline font and weight information. This value can be used as an input to *GetLineFontInfo* and *GetLineFontName*. If this value is -1, then the user has probably modified the line display manually in the drawing and you should use the *LineFont* and *LineWeight* return values to recreate the exact polyline style.
 - *LineFont* = value is used for polyline font information. This value can be used as an input to the *GetLineFontInfo2* and *GetLineFontName2* functions. This value will only be valid if *LineStyle* is -1.
 - *LineWeight* = polyline weight where Thin = 0.0, Normal = 1.0, Thick = 2.0. This value will only be valid if *LineStyle* is -1.
 - *LayerID* = integer value indicating which layer holds this polyline. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::!GetLayerById](#).
 - *LayerOverride* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if *LayerOverride* was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
- next double is the number of points lying on the arc
- next (numPoints * 3) doubles are the coordinates of points lying on the arc

For each ellipse:

- first double gives the type
- next double gives the number of doubles required to hold ellipse geometry (center, unit vector along majorAxis, unit vector along minor axis, major radius, minor radius, startPt, endPt, normal) == 17
- next three doubles give the coordinates of center
- next three doubles give the coordinates of unit vector along major axis
- next three doubles give the coordinates of unit vector along minor axis

- next doubles gives the major radius
- next double gives the minor radius
- next three doubles give the coordinates of startPt
- next three doubles give the coordinates of endPt
- next six doubles are line attributes (see description of line attributes in previous bulleted item)
- next double is the number of points lying on the ellipse
- next (numPoints * 3) doubles are the coordinates of points lying on the ellipse

For each spline

- first double gives the type
- next double gives the number of doubles required to hold the information about the spline geometry (5 + numCtrlPts* numDims + numKnot); if the spline is rational, numDims ==4 otherwise numDims ==3
- next double gives degree = order -1
- next double gives whether the spline is rational (1) or non-rational (0)
- next double gives numDims (i.e., number of coordinates associated with each control point; if the spline is rational , these are four, else three)
- next double gives the number of control points of the spline
- next double gives the number of knots on the spline
- next (numCtrlPoints*numDims) number of doubles are the coordinates of control points; each control point has numDims number of coordinates
- next "numKnots" number of doubles; the knots on the spline
- next six doubles are line attributes (see description of line attributes in previous bulleted item)
- next double is the number of points lying on the spline
- next (numPoints * 3) doubles are the coordinates of points lying on the spline
- For each polyline
 - first double gives the type
 - next double holds value 0.0 as there is no geometry information for a polyline
 - next six doubles are line attributes (see description of line attributes in previous bulleted item)
 - next double is the number of points lying on the polyline
 - next (numPoints * 3) doubles are the coordinates of points lying on the polyline

This method also lets you include or exclude crosshatch lines. This method does not return data for sketch segments added by a user in the drawing view.

To get...	Call...
Size of array needed to hold this data in COM applications	IView::GetPolyLinesandCurvesCount
Items that were sketched in this drawing view	IView::IGetArcs4
	IView::IGetLines4
	IView::IGetSplines3
	IView::IGetEllipses5
If...	Then...
Display mode of the view is: • Shaded • Shaded with Edges • Draft Quality • Fast HLR/HLV	A value is not returned. Use IView::SetDisplayMode3 to change Shaded or Shaded with Edges mode to Wireframe, Hidden Lines Removed (HLR), or Hidden Lines Visible (HLV), and then get the polylines.
Changes are made to the parts or assemblies shown in this drawing	Polylines are only generated that are in the visible viewing bounds when the drawing is opened.

Drawing is already open

All polylines in the drawing are generated. If you open a drawing that is zoomed in to a particular region, then the polylines that are outside the zoomed region do not exist if the parts or assemblies shown in this drawing have been changed. To force the generation of all possible polyline data, call [IModelDoc2::ViewZoomtofit2](#).

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetDisplayMode2 Method](#)
[IView::GetPolyLinesAndCurvesCount Method](#)
[IView::GetPolylines7 Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPolyLinesAndCurves Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetPolyLinesAndCurves Method (IView)

Gets the arcs, ellipses, splines, and polylines in the view with an option to include or exclude crosshatch lines.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPolyLinesAndCurves( _  
    ByVal CrossHatchOption As System.Short, _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IView  
Dim CrossHatchOption As System.Short  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetPolyLinesAndCurves(CrossHatchOption, ArraySize)
```

C#

```
System.double IGetPolyLinesAndCurves(  
    System.short CrossHatchOption,  
    System.int ArraySize  
)
```

C++/CLI

```
System.double IGetPolyLinesAndCurves(  
& System.short CrossHatchOption,  
& System.int ArraySize  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CrossHatchOption

- 0 = include crosshatch lines
- 1 = exclude crosshatch lines
- 2 = include only crosshatch lines

ArraySize

Size of the array of lines and parameters in the view

Return Value

- in-process, unmanaged C++: Pointer to an array of lines and parameters in the view (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

The difference between this method and [IView::GetPolylines6](#) and [IView::!GetPolylines6](#) is that splines and ellipses returned by this method are returned as equation parameters, which are more precise definitions.

This method returns data for all visible solid model edges in the drawing view. This includes these types of edges:

- arcs: Returned using a center point, radius, and start and stop location.
- circles: Returned using a center point, radius, and start and stop location.
- ellipses: Returned as equation parameters.
- splines: Returned as equation parameters.
- straight lines: Returned as tessellated polylines, but because the edges are straight, there is no loss of data in the approximation.
- For each arc:
 - first double gives the type
 - next double gives the number of doubles required to hold arc geometry (center, startPt, endPt, normal) == 12
 - next three doubles give the coordinates of center
 - next three doubles give the coordinates of startPt
 - next three doubles give the coordinates of endPt
 - next three doubles give the coordinates of the normal
 - next six doubles are line attributes
 - *LineColor* = polyline color. This value is determined either by the explicitly set value or by the layer that the polyline is on.
 - *LineStyle* = value combines polyline font and weight information. This value can be used as an input to *GetLineFontInfo* and *GetLineFontName*. If this value is -1, then the user has probably modified the line display manually in the drawing and you should use the *LineFont* and *LineWeight* return values to recreate the exact polyline style.
 - *LineFont* = value is used for polyline font information. This value can be used as an input to the *GetLineFontInfo2* and *GetLineFontName2* functions. This value will only be valid if *LineStyle* is -1.
 - *LineWeight* = polyline weight where Thin = 0.0, Normal = 1.0, Thick = 2.0. This value will only be valid if *LineStyle* is -1.
 - *LayerID* = integer value indicating which layer holds this polyline. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) and [ILayerMgr::!GetLayerById](#).
 - *LayerOverride* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if *LayerOverride* was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
- next double is the number of points lying on the arc
- next (numPoints * 3) doubles are the coordinates of points lying on the arc
- For each ellipse:
 - first double gives the type
 - next double gives the number of doubles required to hold ellipse geometry (center, unit vector along majorAxis, unit vector along minor axis, major radius, minor radius, startPt, endPt, normal) == 17
 - next three doubles give the coordinates of center
 - next three doubles give the coordinates of unit vector along major axis
 - next three doubles give the coordinates of unit vector along minor axis
 - next doubles gives the major radius
 - next double gives the minor radius
 - next three doubles give the coordinates of startPt
 - next three doubles give the coordinates of endPt
 - next six doubles are line attributes
 - next double is the number of points lying on the ellipse
 - next (numPoints * 3) doubles are the coordinates of points lying on the ellipse
- For each spline
 - first double gives the type
 - next double gives the number of doubles required to hold the information about the spline geometry (5 + numCtrlPts* numDims + numKnot); if the spline is rational, numDims ==4 otherwise numDims ==3
 - next double gives degree = order -1
 - next double gives numDims (i.e., number of coordinates associated with each control point; if the spline is rational , these are four, else three)
 - next double gives the number of control points of the spline
 - next double gives the number of knots on the spline

- next (numCtlPoints*numDims) number of doubles are the coordinates of control points; each control point has numDims number of coordinates
- next "numKnots" number of doubles; the knots on the spline
- next six doubles are line attributes
- next double is the number of points lying on the spline
- next (numPoints * 3) doubles are the coordinates of points lying on the spline
- For each polyline
- first double gives the type
- next double holds value 0.0 as there is no geometry information for a polyline
- next six doubles are line attributes
- next double is the number of points lying on the polyline
- next (numPoints * 3) doubles are the coordinates of points lying on the polyline

This method also lets you include or exclude crosshatch lines. This method does not return data for sketch segments added by a user in the drawing view.

To get...	Call...
Size of array needed to hold this data	IView::GetPolyLinesAndCurvesCount
Items that were sketched in this drawing view	IView::GetArcs4 and IView::IGetArcs4 IView::GetLines4 and IView::IGetLines4 IView::GetSplines3 and IView::IGetSplines3 IView::GetEllipses5 and IView::IGetEllipses5

If...	Then...
Display mode of the view is: • Shaded • Shaded with Edges • Draft Quality • Fast HLR/HLV	A value is not returned. Use IView::SetDisplayMode3 to change Shaded or Shaded with Edges mode to Wireframe, Hidden Lines Removed (HLR), or Hidden Lines Visible (HLV), and then get the polylines.
Changes are made to the parts or assemblies shown in this drawing	Polylines are only generated that are in the visible viewing bounds when the drawing is opened.
Drawing is already open	All polylines in the drawing are generated. If you open a drawing that is zoomed in to a particular region, then the polylines that are outside the zoomed region do not exist if the parts or assemblies shown in this drawing have been changed. To force the generation of all possible polyline data, call IModelDoc2::ViewZoomtofit2 .

See Also

[IView Interface](#)
[IView Members](#)
[IView::GetPolyLinesAndCurves Method](#)
[IView::GetPolyLineCount5 Method](#)
[IView::GetDisplayMode2 Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetProjectionArrow Method (IView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IView Interface](#) : IGetProjectionArrow Method (IView)

Gets the projection arrow for this projected view.

.NET Syntax

```
Function IGetProjectionArrow() As ProjectionArrow
```



```
Dim instance As IView  
Dim value As ProjectionArrow  
  
value = instance.IGetProjectionArrow()
```

```
ProjectionArrow IGetProjectionArrow()
```

```
ProjectionArrow^ IGetProjectionArrow();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Projection arrow](#)

Visual Basic for Applications (VBA) Syntax

See [View::IGetProjectionArrow](#).

See Also

[IView Interface](#)
[IView Members](#)
[GetProjectionArrow Method](#)
[IView::GetProjectionLineCount Method](#)
[IView::GetProjectionLines Method](#)
[IView::IGetProjectionLines Method](#)
[IProjectionArrow::IGetProjectedView Method](#)
[IProjectionArrow::GetProjectedView Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0